

Key features

- **Promise-based API:** Axios allows chaining `.then()` and `.catch()` methods for handling asynchronous operations.
- **Request and response interception:** Developers can modify requests or responses before they are sent or received.
- **Easy integration:** Axios integrates seamlessly with other libraries and frameworks like React, Vue, and Angular.

Use cases: This library is commonly used to fetch data from APIs and handle RESTful requests in modern web applications.

Pros: Axios simplifies working with HTTP requests and provides excellent error handling.

Cons: It introduces slight overhead compared to the native Fetch API, though the added functionality may justify the trade-off.



Anime.js

Anime.js is a lightweight animation library for creating smooth and interactive animations. It supports CSS properties, SVG, and JavaScript objects.

Key features

- **Animation support:** Anime.js can animate a wide range of properties, including CSS styles, transforms, and SVG elements.
- **Ease of use:** The API is simple, and the library is lightweight, making it ideal for quick animation tasks.

Use cases: Anime.js is used in websites and applications that require smooth, interactive animations and transitions.

Pros: It is highly customisable and provides powerful animation features with minimal overhead.

Cons: It may not be well-suited for large-scale applications where more complex animation frameworks are needed.

How to choose the right library

Selecting the right JavaScript library is crucial for the success of your project. It involves evaluating several key factors such as project requirements, performance, learning curve, and community support. Understanding these aspects helps you choose a library that aligns with your needs, ensuring faster development, better performance, and ease of use.

Table 1: A comparison of the different JavaScript libraries

Library	Key features	Pros	Cons
React	Virtual DOM, Component-based	High performance, Flexibility	Steep learning curve
jQuery	AJAX, Event handling, Animations	Easy to use, Cross-browser compatible	Less relevant for modern frameworks
D3.js	Data binding, SVG, Canvas support	Precision, Flexibility	Steep learning curve
Vue.js	Two-way data binding, Lightweight	Simple, Modular	Smaller ecosystem than React
Lodash	Utility functions, Performance	Time-saving, Optimised for performance	Modern JS features reducing necessity
Three.js	3D rendering, Animation support	Stunning visuals, Robust API	Performance bottlenecks on low-end devices
Moment.js	Date parsing, Time zone handling	Comprehensive functionality	Considered legacy, Modern alternatives
Chart.js	Predefined charts, Responsiveness	Easy to use, Responsive	Less customisable for complex charts
Axios	Promise-based API, Request handling	Easy syntax, Excellent error handling	Slight overhead over Fetch API
Anime.js	CSS, SVG, JavaScript animation	Lightweight, Powerful animations	Less suited for large-scale applications